

The Problem of Measurement and Control of Speed in a Drive with an Inaccurate Measuring Position Transducer

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Abstract- The theoretical limitations of the speed control for drives equipped with an inaccurate, low-resolution position sensor (encoder) are analysed in this paper. Encoder errors as well as time delay introduced by the speed calculation method are considered. The conclusions arising from the theoretical analysis have been confirmed by simulation results.

I. INTRODUCTION

Speed is not usually measured in modern drives, but is calculated indirectly from the measurement of the angular position or based on the measurements of electric sizes (sensorless drives). It is useful to consider what dynamics of the control of the speed one can reach, when the only measuring signal is the quantization angle position of the shaft. In this paper the drive is considered as ideal, which means that losses, delays, the changeability of parameters and different nonlinearity other than the characters of the measuring transducer are neglected. Only the simplest method for the calculation of the speed was taken into consideration (the method T [1]) and the simple regulators PI / PID. The reaction to the changes of the moment of the load torque m_L is analysed; the reaction to the changes of the reference speed ω_r in the ideal drive can be formed freely, using e.g. a structure with a

pre-filter F [2], shown in Fig. 1.

II. MODELS OF SPEED CONTROL LOOP.

The block diagram of the non-linear, discrete-continuous model of the drive is presented in Fig. 1a. The torque control loop is represented through the (omitted from the drawing) linear proportional module with unit gain. The continuous time integrator, with gain inversely proportional to the moment of inertia J , represents the dynamics of the drive. The model of the angle position transducer (encoder) contains an ideal integrator and non-linear module with a stair-step function (a quantizer). It is initially assumed that all steps have exactly the same width, equal to the resolution quantum of the measurement angle, Θ . The influence of differences in the width of steps is discussed later, in section 5.

The speed is calculated as the quotient of the quantum of the angle Θ by the measured time T_s between successively appearing impulses from the encoder. So the calculated speed ω_m is the average speed for the last period T_s . It is assumed that the digital controller of the speed calculates the value of the reference torque of the drive immediately, in the moments of the changes of the position signal and the calculation of the new speed. The torque of the drive m is kept at a constant level

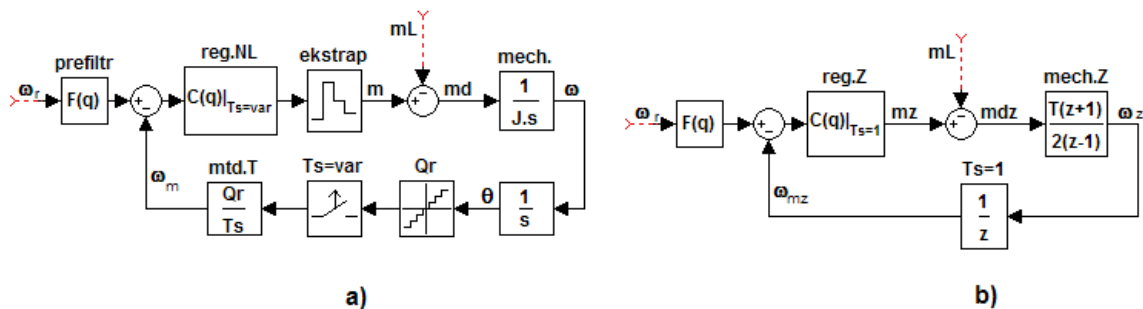


Fig. 1. The diagram of the models of the drive: a) nonlinear hybrid continuous – discrete, with variable sampling time T_s (model NL), b) linearized discrete (model Z) using normalized variables

until appearing the next quantum of the position measurement.

The computational part of the measuring structure and the speed controller can be considered for the discrete-time system, working with the changing, dependent from the speed sampling time T_s . The delay introduced to the loop of the speed control by the measuring structure and the speed controller is also variable and dependent from the current speed. As a consequence, the settings of the controller should also depend on the speed.

The described non-linear model of the drive can easily be studied by simulation. In order to obtain simple approximate examples of the controller settings and the reachable quality index of the control, the model should be linearized by using very small variations of the speed from the steady state value Ω_0 . The distance T_s between the neighbouring quanta are then almost the same, which lets the whole system be treated as a discrete-time system with constant sampling $T_s = \Theta_r / \Omega_0$ (Fig. 1b). The continuous integrator in the model of the mechanical part becomes replaced by a discrete integrator with Heun's formula (trapezoid rule), and the whole system of the position measurement and the speed calculation is by simple unit delay with T_s . The substitute delay T_s consists on delay equal $T_s / 2$ resulting from the calculation of the average speed for the past period, and the delay introduced by the extrapolator on the controller output

The following base units were introduced for the simplification and generalization of the analysis:

- $\Omega_B = |\Omega_0|$ - base speed equal to absolute value of steady state drive speed,
- $\Theta_B = \Theta_r$ - base angle equal to resolution quantum of encoder,
- $T_B = T_s = \Theta_B / \Omega_B$ - base time equal to period between successive changes of measured position,
- $A_B = \Omega_B / T_B = \Omega_B^2 / \Theta_B$ - base angular acceleration,
- $M_B = J \cdot A_B = J \cdot \Omega_B^2 / \Theta_B$ - base torque.

In the normalized model coefficients J , T_s , and Ω_0 are not present. In order to interpret the results in physical units, the normalized variables should be multiplied by the suitable base units. The normalized time constants are multiplied by T_B , and gains are multiplied by following base gain:

$$K_B = M_B / \Omega_B = J / T_B = J \cdot \Omega_0 / \Theta_r. \quad (1)$$

According to the described principles simulating models were built in the Matlab/Simulink system, denominated as NL (non-linear) and Z (linear, discrete).

III. THE TUNING OF THE CONTROLLER AND QUALITY INDEX ON THE STEP CHANGE OF LOAD TORQUE

The linear, discrete PID controller is taken into consideration. The transfer function of such a block can be written as (2), where $q = z^{-1}$ is the delay operator:

$$C(q) = K_p \left(1 + \frac{T_s}{T_i} \frac{1}{1-q} + \frac{T_d}{T_s} (1-q) \right) \quad (2)$$

In equation (2), the same coefficients are present as in a similar continuous controller: K_p - the proportional gain, T_i - the integration time constant, and T_d - the differentiating time constant. To obtain the generality in formula (2), there is the sampling time T_s , but in normalized models considered later it has the unit value $T_s=1$.

The rough tuning of the PI controller can be do on the basis of a simple formula, linking to the symmetry criterion, on the assumption that the substitute small time constant is equal to the delay T_s and it has the relative value equal to 1.

In such a case, the controller settings are as follows (denominated as PI-1):

$$K_p = 0.5; \quad T_i = 4.0; \quad T_d = 0 \quad (3)$$

In Fig. 2 a, c, e and g the response of the speed to the step change of load torque Δm_L for a system with speed controller tuning according to the PI-1 formula is presented.

The following characteristic parameters were taken on in the valuation of responses:

$t_{u2\%} / T_s$ - the setting time of the speed response with a 2 % accuracy region

$\Delta \omega_{max} / \Delta m_L$ - the maximal dynamic speed error

$\Theta_u / \Delta m_L$ - the integral of speed error (the angular error).

In the case of linear model Z all responses are similar to each other: only the speed scale changes. The PI controller can be optimized by using various criteria.

When the criterion of the minimum of speed setting time was applied, then the following controller setting were obtained (denominated as PI-2):

$$K_p = 0.55; \quad T_i = 3.65; \quad T_d = 0 \quad (4)$$

The PID controller's rough set-up, using the rules of Ziegler-Nichols or Chien Hron-Reswick [3] gives strong oscillative answers. Because of this, the speed setting time is quite long, despite the significant decrease of the maximum speed error and the integral of speed error. When the criterion of the minimum of speed setting time was applied to the PID controller, then the following controller settings were obtained:

$$K_p = 0.7; \quad T_i = 2.6; \quad T_d = 0.3 \quad (5)$$

In Fig. 2 b, d, f and h the response of the speed to the step change of load torque Δm_L for a system with speed controller tuning according to the presented PID formula is shown.

The summary list of the response parameters is shown in Table 1.

Linear model Z approach the operation of non-linear model NL closely, when the speed changes imperceptibly from the steady state value. When this condition is not fulfilled, the results obtained from linear model Z are too optimistic - the real drop of the speed and the setting time can be considerably larger. In Fig. 2 c and f is shown the speed response to the large step change of the load torque $\Delta m_L=0.3$.

Differences become very clear when the speed falls about 50% below the steady state speed, and the period T_s lengthens above twice. When the speed is larger than the speed set (e.g. after applying the negative load torque - breaking mode,

Table 1. Response parameters for PI/PID controllers.

Quality Index	Model								
	Linear, discrete Z			Nonlinear hybrid NL					
				$\Delta m_L = 0,01$			$\Delta m_L = 0,1$		
	PI-1	PI-2	PID	PI-1	PI-2	PID	PI-1	PI-2	PID
$t_{u2\%} / T_s$	9	7	5	9.1	7.24	5.4	12.6	12.9	7.9
$\Delta \omega_{\max} / \Delta m_L$	1.74	1.65	1.2	1.82	1.73	1.37	1.95	1.84	1.46
$\Theta_u / \Delta m_L$	8	6.64	3.71	8.1	6.72	3.75	9.2	7.59	4.13

$\Delta m_L = -0.3$, Fig. 2 g h)- worsening of the dynamics of the control because of non-linearity is only insignificant.

IV. THE INTERPRETATION OF RESULTS IN REAL UNITS

Numbers passed as the coefficients of the gain of the controller K_p should be multiplied by K_B (1). This base gain depends on the changing current steady state speed $|\Omega_0|$. Because this value is known with the time lag – we propose to use the values of the reference speed $|\omega_r|$ to tune the controller. This can be done based on the assumption, that the reference speed will not change too quickly and that the area of the near zero speed will be switched off. The time constants T_i and T_d of the controller transmit function are related to changeable sampling time T_s and they do not require additional tuning in the function of the speed.

The numbers passed in the previous section, as the setting times should interpret as the number of periods T_s . The setting times, like the time constants, are then proportional to the sampling period $T_s = \Theta_r / |\Omega_0|$, changing with the speed. For large speeds and the precise measurement of the angle (the small quantum of the resolution Θ_r), the time T_s can be so small (a few μs) that the assumptions about the immediate (in the relation to T_s) working of the torque control loop and making the calculations of the speed controller become unreal. In such a case, the controller should be tuned according to real delays in the whole system.

Introduced formulas are, however, useful prior to using an encoder with low precision, then there is the large quantum of the resolution Θ_r for motion with low rotator speed. For near zero speeds the time T_s aims to infinity, which means that the control loop of the speed can not work. The speed controller acts as an open loop control, or additional signals should be use for the calculation of the speed (observer-based sensorless control).

On the basis of the presented analysis, the minimum speed can define, for which the speed control loop of the drive can still work satisfactorily. The simple method depends on checking the attainable setting time of the speed response to the small disturbance of the load torque.

$$t_{u2\%} [s] = \left(t_{u2\%} / T_s \right) \cdot T_B = \left(t_{u2\%} / T_s \right) \cdot \frac{\Theta_r}{\Omega_0} = \left(t_{u2\%} / T_s \right) \cdot \frac{2\pi N}{\Omega_0} \quad (6)$$

where N is the number of impulse per revolution.

For the low-resolution transducer ($N=48$), a setting times below 100 ms can be reach if the speed is not lower than 1 rev/s, and the best PID controller is used. For a PI controller, the bottom limit of the speed is about twice as high as for the PID controller. The different way of finding the minimum speed is based on checking, if a small step change of load torque $\Delta m_L = a \cdot M_N$ is applied, that will not cause a speed drop larger than the set value $\Delta \omega < b \cdot \Omega$. In the calculations, example data were taken for the drive system ($M_N = 1.5 \text{ N}\cdot\text{m}$, $N=48$, $J=0.00087 \text{ kg}\cdot\text{m}^2$), and the following coefficients were taken: $a=0.1$ and $b=0.25$. The proper PI controller was selected.

In such a case, the smallest speed $\Omega_{0\min}$ for which the described criterion is kept is 2 rev/s.

$$\Omega_{0\min} = \sqrt{\left(\Delta \omega_{\max} / \Delta m_L \right) \cdot \frac{a}{b} \cdot \frac{M_N}{J} \cdot \frac{2\pi}{N}} = 2 [\text{rev/s}] \quad (7)$$

The introduced formula makes possible a fast estimation of the available range of the speed control

V. THE INFLUENCE OF THE ERROR MEASUREMENT ANGLE ON THE SPEED CONTROL LOOP

In the real measurement system, especially if a low-resolution sensor is used, it should matter that considerable error manifests itself in the various widths of the quantizer steps. The preliminary analysis of the influence of these mistakes on the speed control loop can be done using the linear model Z. The additional assumption is that only the single reading of the position is inaccurate, i.e. it is burdened the error $\Delta \Theta_r$.

This will cause two successive results of the calculations of the speed ω_m to be incorrect and these errors differ from each other only in the sign, and their absolute value will be equal to $\Delta \Theta_r / T_s$. This case can be analysed in the linear model Z (Fig 1. b), if the impulse disturbance signal is used instead of reference speed ω_r , and the pre-filter has a transfer function like to differentiating blocks $F(q)=1-q$.

In Fig. 3 are shown the courses of the response torque Δm_z and speed $\Delta \omega_z$ to this type of disturbance with the unit value ($\Delta \Theta_r = 1$), for two previously analysed controllers: PI and PID. The value of the response of the torque signal Δm_z in the initial moment is proportional to the gain of the controller. The subsequent course of the response also depends on all parameters set of the controller. To obtain the physical values

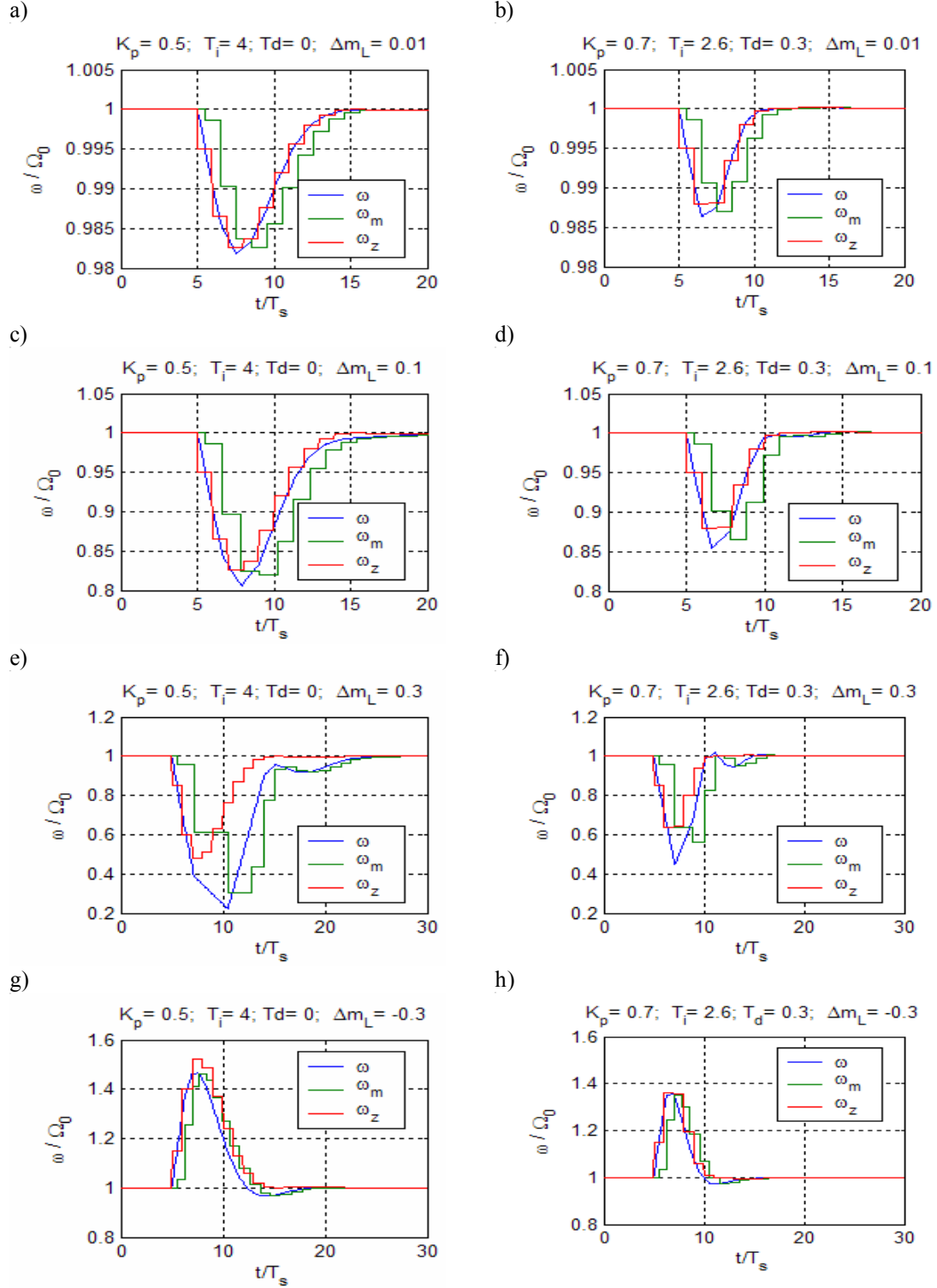


Fig. 2. Normalized response of speed for a drive with a PI controller (a, c, e, g) and a PID controller (b, d, f, h) to step change of load torque Δm_L ; ω : speed from mode NL, ω_m : average speed from model NL, ω_z : speed from model Z

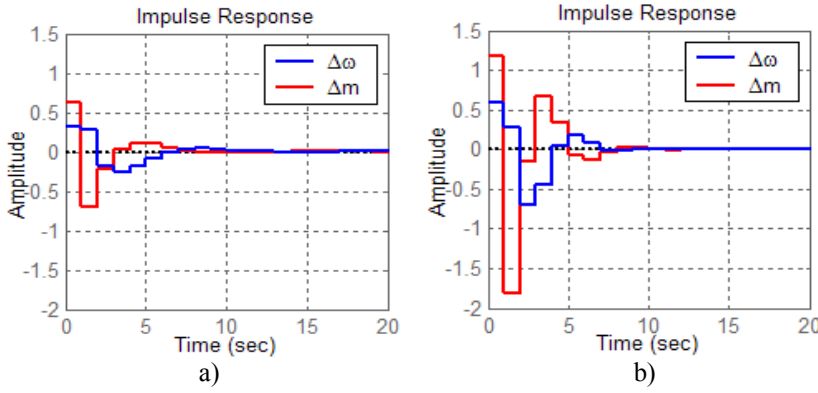


Fig. 3 Response for linear model Z ($T_s=1$) on impulse unitary signal ω_r , with differentiating pre-filter
a) PI controller b) PID controller

the scale of time, torque and speed should be recalculated in the following way

$$t[s] = t \cdot T_s = t \cdot \frac{\Theta_r}{\Omega_B} \quad \Delta m [\text{Nm}] = \delta_\Theta \cdot \Delta m \cdot M_B$$

$$\Delta \omega [\text{rad/s}] = \delta_\Theta \cdot \Delta \omega \cdot \Omega_B \quad (8)$$

where: $\delta_\Theta = \Delta \Theta_r / \Theta_r$ is the real relative transducer error, and t , Δm and $\Delta \omega$ – are normalized values read from the Fig. 3.

The perturbation of the speed, and especially the torque, caused by the error of the encoder, grow with the growth of the steady state speed Ω_0 and can easily reach an inadmissible level. Assuming the maximum admissible pulsation level of the torque $\Delta m_{\text{adm}} = c \cdot M_N$, and knowing the maximum perturbation Δm_{max} of the normalized torque from the impulse response of model Z, one can calculate the range of the speed Ω_0 , which will not come to the excessive pulsation of the torque:

$$|\Omega_0| \leq \sqrt{\frac{c \cdot M_N \cdot 2\pi / N}{\delta_\Theta \cdot \Delta m_{\text{max}} \cdot J}} \quad (9)$$

Assuming the relative error of the encoder $\delta_\Theta = 0.1$ and admissible torque pulsation coefficient $c = 0.1$ and using the remaining data such as in the previous example, one gets the result: $|\Omega_0| < 3 [\text{rev/s}]$. These results confirm that the inaccuracies of the transducer can make up an extraordinarily important regulating problem.

For the described example, the PI controller would give a satisfying result only for the narrow range of speed between 2 and 3 rev/s.

VI. CONCLUSION

Low-resolution encoder errors (inaccuracy as well as quantization error) considerably influence the speed control loop. The feedback speed control signal, calculated from

discrete changes in measured position, can be obtained after a variable time delay, which depends on the encoder resolution and the average speed of the drive. These factors limit the low-speed operation of the drive. The formula for the lowest speed, based on admissible dynamic error, has been derived for PI/PID-controlled drives. Encoder inaccuracy causes torque pulsation, which can be excessive for higher rotational speeds. Some proposals to solve this problem partially can be summarized as follows:

- The repeatable encoder errors can be identified by experiment and memorized, then the proper able of position corrections can be used in the speed calculation method;
- Instead of applying an extra filter, the speed calculation and control procedures can be activated infrequently, every n -th pulse of the encoder;
- The speed signal calculated from the encoder position can be filtered by a low-pass filter, and the speed controller redesigned for the purpose of achieving a compromise between torque pulsation and dynamic errors.

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